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MECHANISTIC INTERPRETABILITY · AGENTIC AI

Opening the Black Box: Mechanistic Interpretability of Agent Tool Selection with Sparse Autoencoders

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Mechanistic
Interpretability
for AI Agents

subject Nemotron-3-Nano-30B
gen-label-judge Qwen3-VL-235B
compute NVIDIA · 4×GB200

KEY RESULT — CAUSAL, NOT JUST CORRELATED

10.1%

of agent tool choices reverse when 10 SAE features are ablated.

Removing a handful of features **causally changes** the agent's tool choice — controls stay flat at zero.

random ablation 0%

SAE round-trip 0%

p = 0.002

also significant: single vs. multi-tool — 17.1% vs. 2.4% baseline (n = 41, p = 0.029).

fundamental vs. technical analysis (finance domain, 1 of 23 contrast types tested) — Nemotron-3-Nano-30B (52 layers), probed at layer 20.

01 The Tool-Selection Black Box

Modern AI agents autonomously pick among databases, code, web search, and sub-agents. **When** the agent chooses is observable — **why** it chose remains opaque.

- Prompt engineering — correlational, not causal.
- Behavioural testing — input/output only.
- Chain-of-thought — plausible, often confabulated.

We open the black box at the token where the model **commits to a tool**, and decompose its hidden state into human-readable features.

02 Pipeline

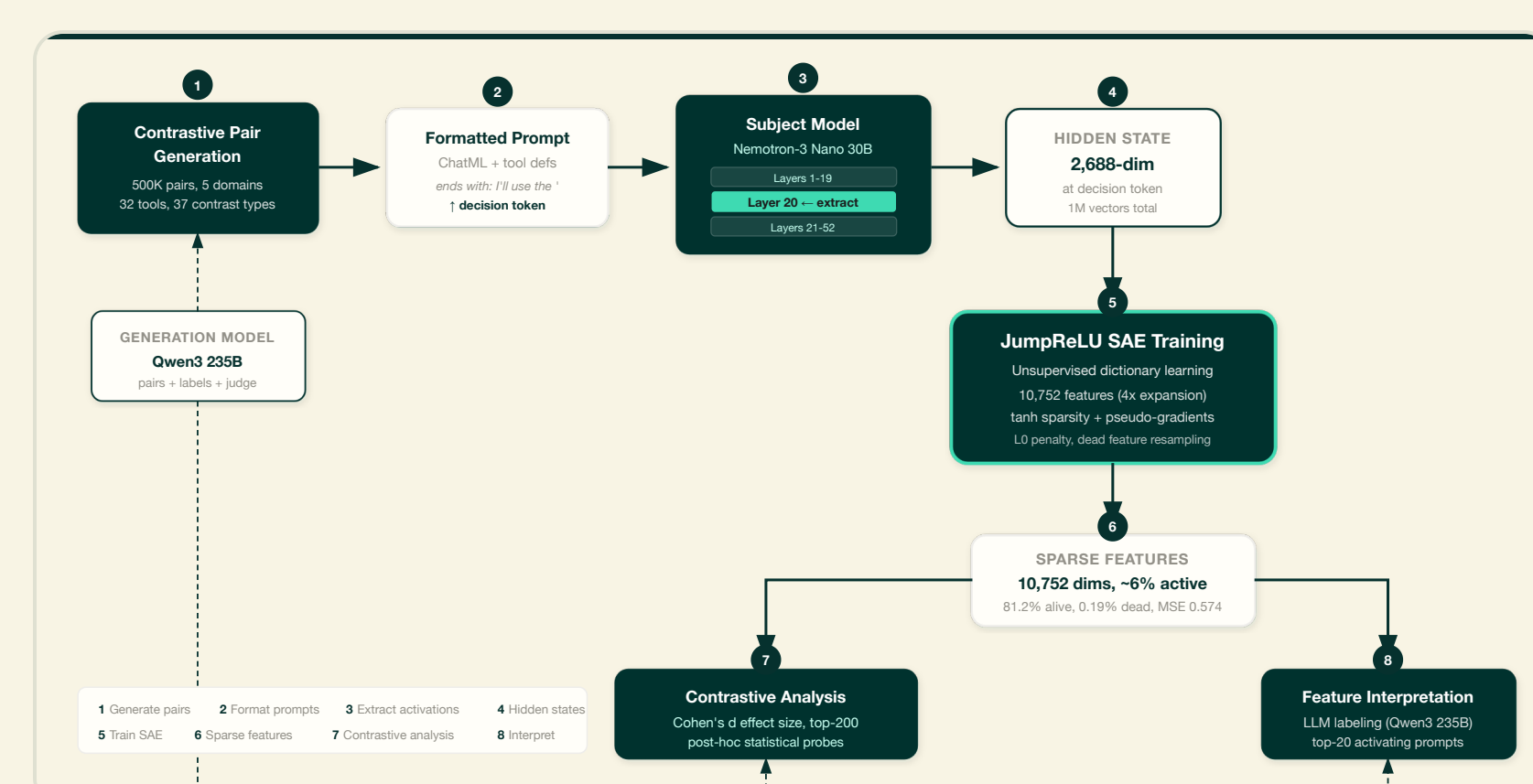


Fig 1. The SAE is trained **unsupervised** on raw activations. Contrastive pairs enter only at the post-hoc probe (Cohen's d) — never as a training signal.

03 Pick the Right Layer — Not the MoE Ones

A single SAE cannot decompose MoE layers. Reconstruction MSE explodes by **>500×** at the sparse-rousted expert layers.

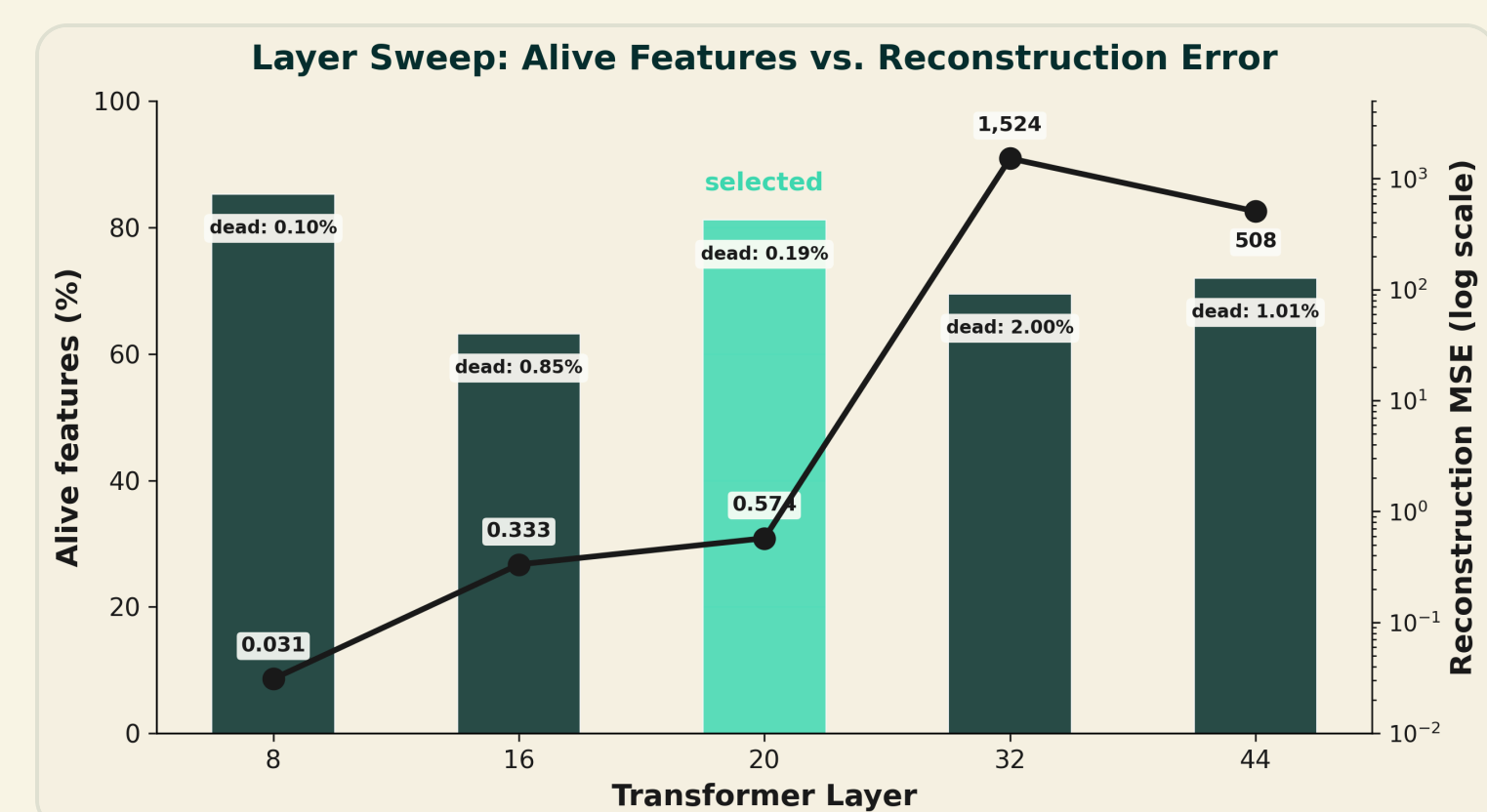
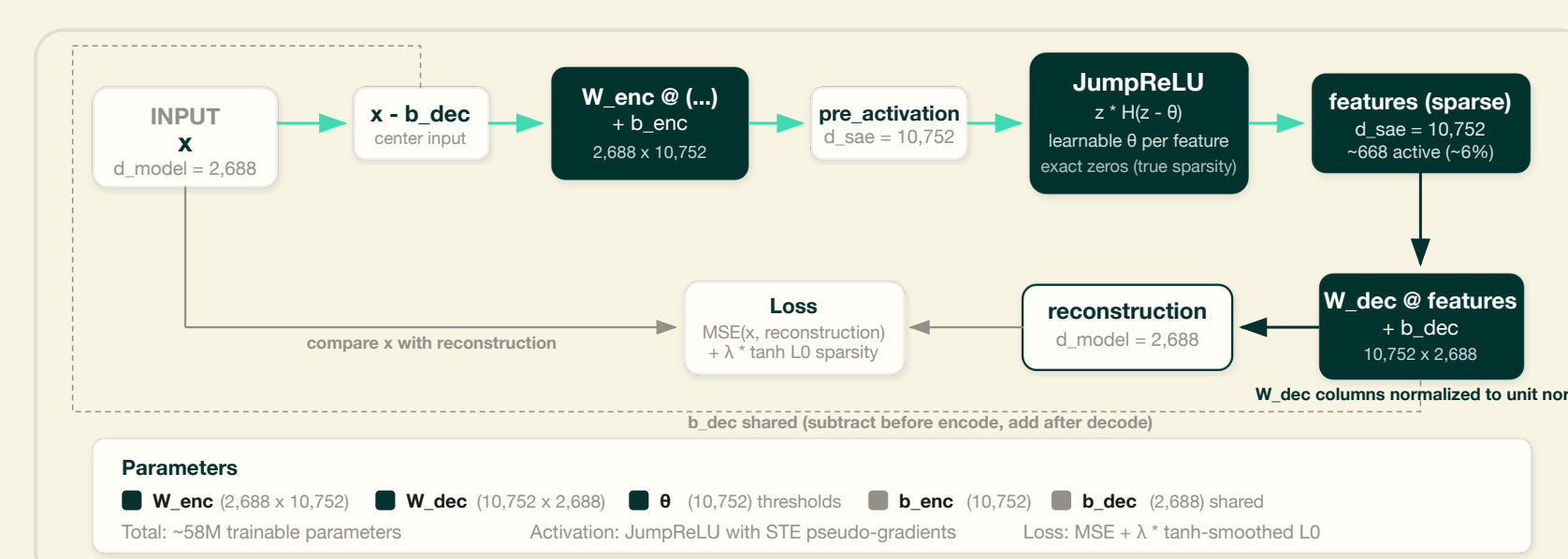


Fig 2. Bars = alive features; line = reconstruction MSE (log). At the MoE layers MSE jumps to **1,524** (L32) and **508** (L44) — vs. **0.574** at layer 20. Layer 20 sits just before the MoE stack with **81.2% alive features**.

04 JumpReLU Sparse Autoencoder

An overcomplete dictionary that decomposes hidden states into **monosemantic** features. JumpReLU thresholds θ_j produce exact zeros — true sparsity — trained with a smooth tanh surrogate.



Dictionary size M	10,752 (4× hidden)
Activations trained on	1,000,000
Reconstruction MSE	0.574
Alive features	81.2%
Active per input	668 ± 26 (6.2%)

Fig 3. Encode → JumpReLU (exact zeros) → decode, with shared b_{dec} and unit-norm W_{dec} . ~58M trainable parameters; loss = MSE + tanh-smoothed L0.

05 Inside a Feature — Emergent, Unsupervised Concepts

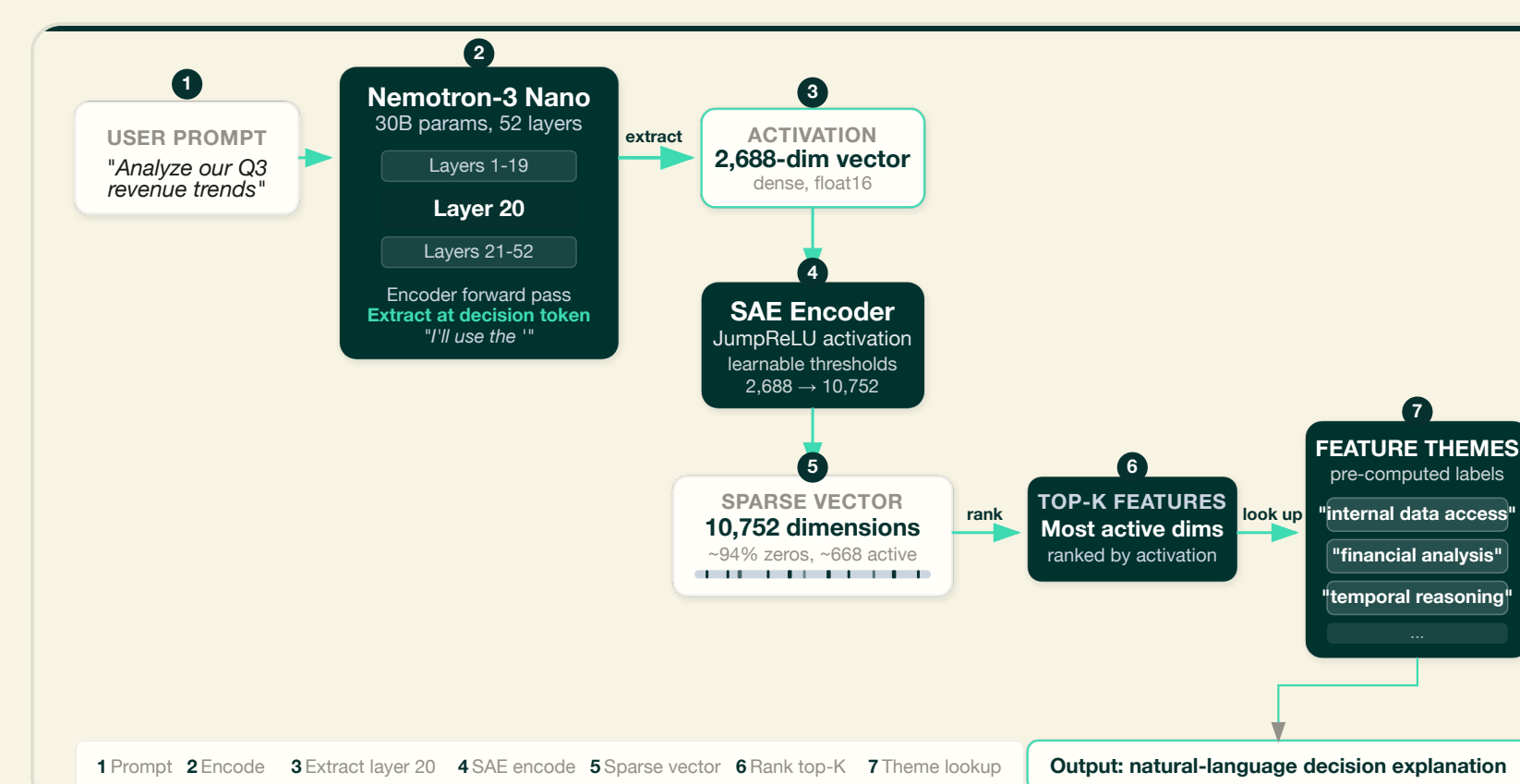
The SAE decomposes Layer-20 activations into **10,752 features**. Without contrastive supervision, named concepts emerge from raw activations alone:

- “internal knowledge retrieval”
- “data modification intent”
- “query complexity”

Each feature carries an **LLM-generated label**, a token-level fuzzing score (aggregate 0.912 ± 0.008 , $p < 10^{-4}$), and its row in the contrastive-ablation table.

06 Strong Focus on Production

Every tool call ships with a rationale. At inference, the top-K active features at the decision token map to pre-computed labels — **no extra LLM call**.



✓ human-readable explanation shipped with every tool call

Fig 4. Top-K active features map to pre-computed theme labels at the decision token — a debugging surface for deployed agents and a foundation for anomaly detection.

07 Feature Label Quality

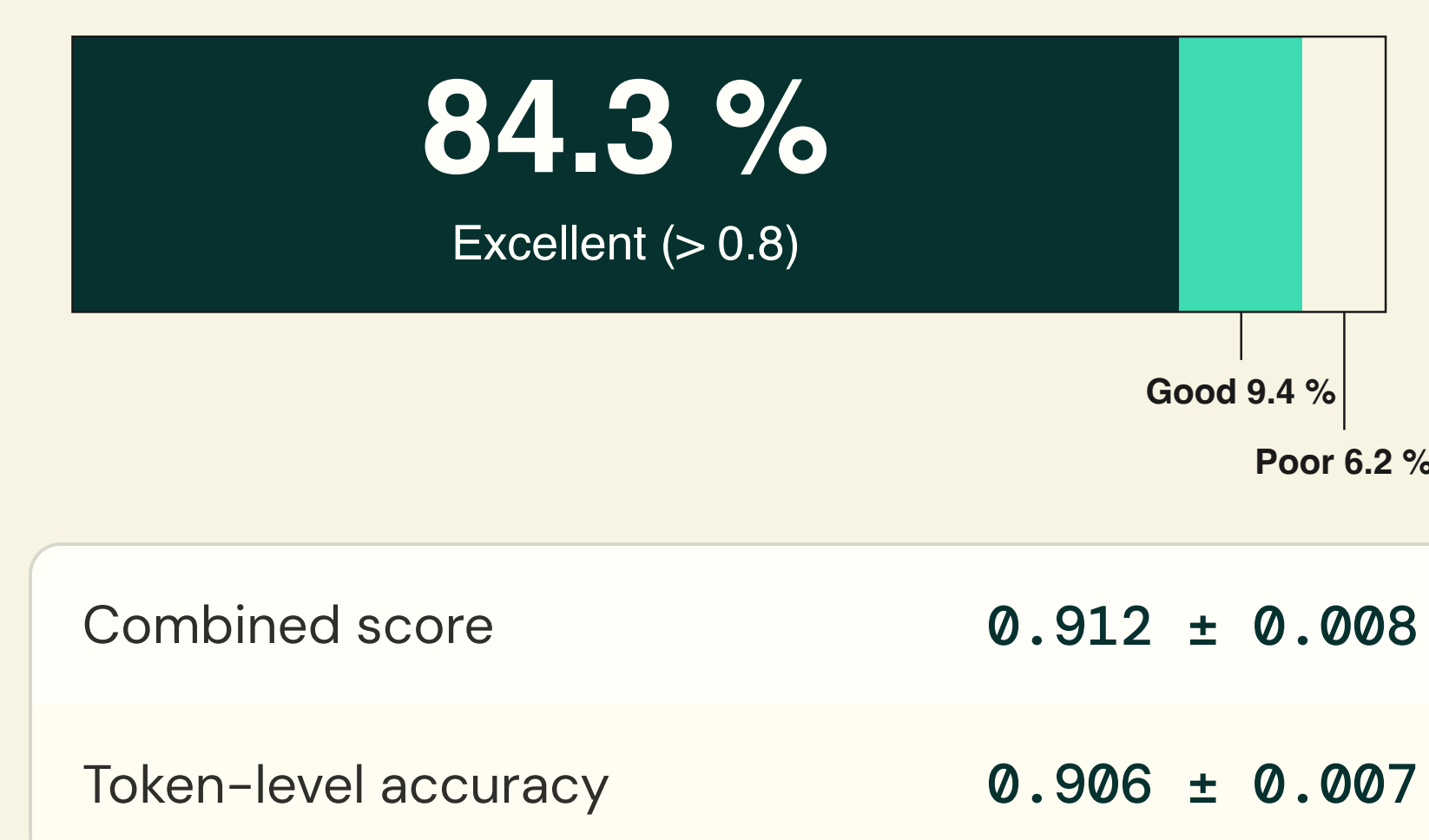


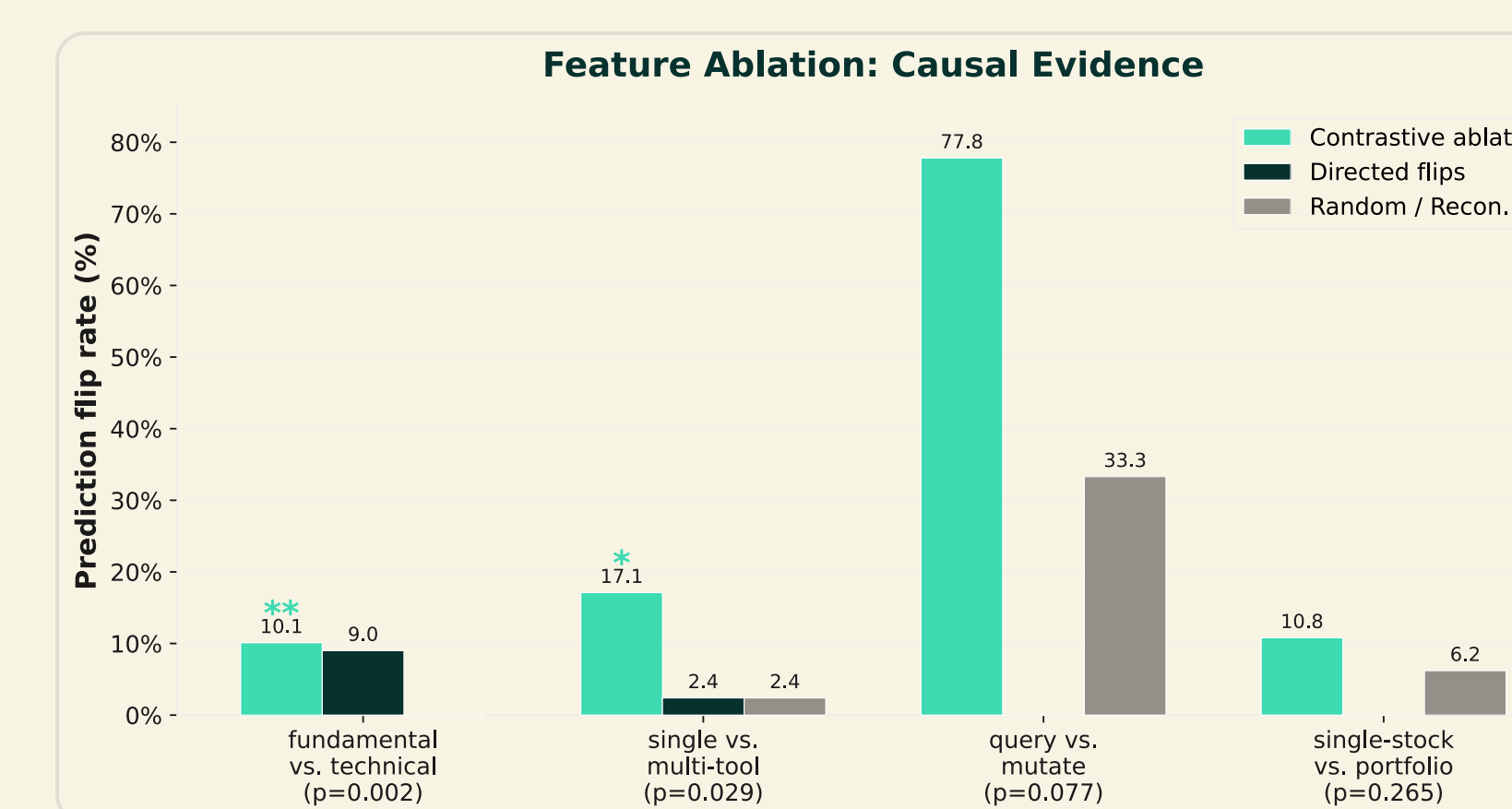
Fig 5. 402 features · 4,422 A/B examples; combined score weights token-level 0.7 vs. prompt-level 0.3 ($p < 10^{-4}$ vs. 0.5). Token-level evaluation catches labels right for the wrong reasons.

Scope & Limitations

- One model, one layer, one decision-token convention.
- Top-10 ablation is conservative; distributed decisions need larger sets.
- Feature labels rely on Qwen3-VL-235B (generation, labelling, and judge) — its errors propagate to rationales.

08 Causal Ablation

Prediction-flip rate by contrast type — significance shown per group.



- Works: fundamental vs. technical — 10.1% flip, controls 0% (p = 0.002).
- Works: single vs. multi-tool — 17.1% vs. 2.4% baseline (n = 41, p = 0.029).
- Null: 9 of 23 contrast types show zero flip — encoded distributively, not in a sparse circuit.

Why it's real: random-ablation flip equals SAE round-trip distortion across all 23 types. The contrastive flip sits **above this noise floor** — it comes from the specific features removed, not encode/decode artefacts.

Code & Paper

github.com/dataiku/kiji-inspector

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COMPUTE NVIDIA · 4×GB200

Prior work

- [1] Bricken, T., Templeton, A., Batson, J., et al. *Towards Monosemanticity: Decomposing Language Models with Dictionary Learning*. Anthropic (2023).
- [2] Rajamanoharan, S., Conmy, A., Smith, L., et al. *Jumping Ahead: Improving Reconstruction Fidelity with JumpReLU Sparse Autoencoders*. arXiv:2407.14435 (2024).
- [3] Juang, C., Paulo, G., Drori, J., Belrose, N. *Open-Source Automated Interpretability for Sparse Autoencoder Features*. EleutherAI Blog (2024).